

Practice Exam Operating Systems August 2019

1 Caching

Given is a cache characterized by a cache size (power of 2), block size (power of 2), associativity (power of 2), and replacement policy (LRU or Random). We can access this cache using the interface:

```
int access_cache(unsigned long InputAddress)
```

The return value is 1 if the operation was a hit, otherwise 0. Each call will also ensure that the requested word from memory (including all other words from the same memory block) is placed in the cache.

Assume we test the cache with the code below. In this code, the parameter cache size is an input parameter that contains the size of the cache.

```
int find(int cache_size) {
    int guess = 0;
    bool found = false;
    access_cache(0);
    while(!found) {
        guess++;
        access_cache(guess*cache_size);
        for(int i = 0; i < guess; i++) {
            if(access_cache(i*cache_size) == 0) {
                found = true;
            }
        }
    }
    return guess;
}
```

1. What is being computed? Explain the reasoning and state which assumptions are necessary for this code to work correctly. (2 pt)
2. Can this code also determine the replacement policy? If adjustments are necessary, explain how you would make these adjustments. Explain how the code, with or without modification, can yield a correct result. (2 pt)

Solution: The associativity is determined. The cache must initially be empty. All addresses that are a cache size apart will always map to the same set. This code fills the first set and checks when blocks are evicted.

You always know that Direct Mapping is used if the associativity equals 1. If the oldest block is not evicted, you know for sure that Random Replacement is used. You can never be certain whether it is LRU, since Random Replacement may sometimes evict the first block, but you can keep adding blocks and see if it consistently evicts the oldest block in the set (though this is not a perfect solution).

```

if(access_cache(i*cache_size) == 0) {
    if(i != 0) {
        bool random_replacement = true;
    }
    found = true;
}

```

2 Virtual Memory

1. We consider a system with a local replacement policy with fixed allocation of 4 frames per process. Assume a process requests the following pages in order: 2, 5, 1, 3, 2, 1, 8, 3, 8, 1, 4, 0. For the Segmented-FIFO replacement algorithm (with each queue containing 2 elements), indicate which pages are in memory after each request. (2 pt)
2. Give one example where the LRU algorithm performs better than the FIFO algorithm and one where the reverse is true. (2 pt)

Solution:

t	Virtual Page	S-FIFO (mirrored)
0	2	2 —
1	5	5 2 —
2	1	1 5 — 2
3	3	3 1 — 5 2
4	2	2 3 — 1 5
5	1	1 2 — 3 5
6	8	8 1 — 2 3
7	3	3 8 — 1 2
8	8	3 8 — 1 2
9	1	1 3 — 8 2
10	4	4 1 — 3 8
11	0	0 4 — 1 3

- The correct sequence earns 2 points; a correct interpretation of the algorithm but a small mistake earns 1 point.

- 1 point per correct example. Example where LRU is better than FIFO (assuming 3 frames): 2 3 1 2 4 2 (4 misses for LRU, 5 for FIFO), FIFO better than LRU: 2 3 1 2 4 3 (4 misses for FIFO, 5 for LRU).

3 Disk Management

Given a system with 6 hard drives. Each hard drive has a storage capacity of 4TB. Assume we configure them with RAID 1+0.

1. State the total storage capacity available to a user of this system. (0.5 pt)

Solution: There are 3 pairs (in RAID 0) of 2 hard drives (in RAID 1). We lose 1 drive per pair due to mirroring. Thus, we have $3 \times 4\text{TB} = 12\text{TB}$ left for the user.

2. How many hard drives can fail before data is lost? (0.5 pt)

Solution: Per RAID 1 pair, 1 drive can fail (the mirror will compensate). In total, up to 3 drives may fail.

3. Now assume we configure the drives with RAID 5+0. How do your answers to the previous two questions change? (1 pt)

Solution: There are now 2 RAID 5 systems, each with 3 drives. The storage capacity is now $4 \times 4\text{TB} = 16\text{TB}$, since 1 drive's capacity is used for distributed parity data per RAID 5 configuration.

RAID 5 provides 1-failure redundancy, so 1 drive per RAID 5 configuration can fail (total of 2 drives).

Explain briefly how you arrive at your answer each time.

4 Scheduling

1. Given a list of N jobs (all available at time 0), each with length p_j , deadline d_j , and weight w_j . Assume $\sum_j w_j = 1$. For a schedule S , C_j is the completion time of job j . Design an algorithm that produces the optimal schedule so that the weighted maximum lateness $\max_j w_j(C_j - d_j)$ is minimized. (2 pt)

Solution: The algorithm works backwards. You calculate the completion time of the last job by summing all job lengths. Then select job j that minimizes $w_j(C_j - d_j)$. If multiple jobs give the same value, select the longest job (this lowers the completion time for the next iteration). Repeat these steps for the remaining jobs until the list is empty.

```
scheduled_list = scheduler(job_list)
    if only 1 job left
        return job
```

```
        else
            completion_time = sum of all job lengths in the job_list
            # (in case of ties)
            last_job = longest job j for which  $w_j(C_j - d_j)$  is minimal
            return append(scheduler(job_list w/o last_job), last_job)
        end
    end
end
```

2. Consider a preemptive scheduling algorithm based on dynamically changing priorities, where higher numbers indicate higher priority. If a process is waiting, its priority increases at rate α ; if it is running, it increases at rate β . New processes receive priority 0.

(a) Which scheduling algorithm does this correspond to if $\beta > \alpha > 0$? (1 pt)

Solution: FIFO/FCFS

(b) Which scheduling algorithm does this correspond to if $\alpha < \beta < 0$? (1 pt)

Solution: LIFO

5 File Systems

Given a file system that uses linked allocation and a linked list to keep track of free blocks.

1. What are the advantages and disadvantages of keeping the linked list of free blocks sorted? (1 pt)
2. Which advantages or disadvantages dominate in a static system (many reads, few writes)? And in a dynamic system (many allocations/deallocations)? Explain briefly. (1 pt)

Solution:

Advantage: when creating files, allocated blocks are closer together. As a result, sequential reading and writing is more performant (ignoring deallocation).

Disadvantage: when deallocating blocks, the freed blocks must be inserted into the list at the correct position to keep it sorted. This takes more time compared to adding them at the start of the list.

In a static system, the advantage outweighs the disadvantage; in a dynamic system, the opposite is true.

(This document has been translated using ChatGPT, the TA has verified the questions but if anything seems wrong, let us know!)

Name:

GUIDELINES

- Begin met het schrijven van je naam op elke pagina én kladblad.
- Dit is een gesloten boek examen.
- Zorg dat je mobiele telefoon uit staat en leg het vooraan in het lokaal.
- Wees zo compleet mogelijk in je antwoorden. Motiveer duidelijk waarom je dit antwoord geeft.
Meestal zijn antwoorden in 1 zin niet voldoende!
- Alle vormen van communicatie zijn verboden.

=====

- Start by writing your name on each page and the additional pages
- This is a closed book exam.
- Make sure your mobile phone is switched off and place it at the front of the room.
- Be as complete as possible in your answers. Motivate clearly why you provide this answer.
Usually answers in 1 sentence are not enough!
- All forms of communication are prohibited.

QUESTION 1: PROCESSOR AND THREADS

- A) Describe three differences that a system using multiple processes will possess when compared with a system using multiple threads.

QUESTION 2: CACHE MEMORY

- A) Describe the Set Associative Mapping technique. Indicate what are the trade-offs that can be encountered depending on how the number of lines per set (k-value) is selected.

QUESTION 3: MEMORY MANAGEMENT

- A) Describe the Dynamic Partitioning method. Indicate what are the advantages of this method over standard fixed partitioning methods. Indicate what potential issues may be encountered by adopting a dynamic partitioning approach.

QUESTION 4: VIRTUAL MEMORY

- A) Describe the organization of Multilevel Page Tables and indicate step by step how mapping between virtual addresses and physical addresses (in main memory) occurs.

Theory Exam - January 20th 2022

Name:

QUESTION 5: PROCESSOR SCHEDULING

- A) Describe the short, medium and long -term scheduling operations that take place during the lifetime of a process. Indicate the moment when they take place, their function and elements/components involved in them.
- B) Describe the operation of Multilevel Feedback Queues for scheduling. Indicate how processes of different length (short vs. long processes) will behave under such scheduling scheme and whether this scheme will favor any of them.

QUESTION 6: DISK MANAGEMENT

- A) Describe how the SCAN and LOOK algorithms operate. Draw a comparison between these two algorithms. In addition, indicate how the cyclic versions (C-SCAN and C-LOOK) of these algorithms addressed the weaknesses present in their original versions (SCAN and LOOK).

QUESTION 7: FILESYSTEMS

- A) Describe what we mean by linked (or chained) allocation in the context of filesystems. Further elaborate on what are its advantages and disadvantages. Indicate whether there are design decisions that can be made in order to address the disadvantages (if any).